

Department of Computer Science and Engineering(CSE)

INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG(IIUC)

Course code:CSE-3524

Course Title: Microprocessor & Microcontroller Sessional

EXPERIMENT: 1

Experiment Name : Familiarization with MDA 8086 Microprocessor kit and its operation in ‘machine codes’.

Objectives :

1. To familiarize with MDA-8086 system configuration
2. To operate MDA-8086 for ‘machine code’ in kit mode.

Familiarization :

The kit is holding these things :

1. CPU (Central processing unit) : Using Intel 8086, Using 4.9152Mhz.
2. ROM (Read Only Memory) : It has program to control user's key input, LCD display, user's program. 64K Byte, it has data communication program. Range of ROM Address is F0000~FFFFFH.
3. SRAM (Static Random Access Memory) : Input user's program & data. Address of memory is 00000H~0FFFFH, totally 64 KByte .
4. DISPLAY : It is LCD, 16(Character) × 2(Line).
5. KEY BOARD : It is used to input machine language and has 16 of hexadecimal keys and 8 of function keys.
6. SPEAKER : Able to test sound using with speaker and further more able to test synthesizer.

7. RS-232C : It is ready to do data communication with IBM compatible personal computer.
 8. DOT MATRIX LED : To understand & test of dot matrix structure and principle of display it is interfaced to 8255A (PPI).
 9. A /D CONVERTER : Convert analog signal to digital signal using with ADC0804.
 10. D /A CONVERTER : Convert digital signal to analog signal using with DAC0800 and it is interfaced so as to more Level meter.
 11. STEPPING MOTOR INTER FACE : So as to control stepping motor driver circuit of stepping motor is interfaced.
 12. POWER : AC 110~220V , DC +5V 3A , +12V 1A , -12V 0.5A SMPS.
- The keyboard has 26 button including reset key. Some for operation and some for input only.

Equipments:

1. MDA-8086 kit,
2. Power supply,
3. Emulator 8086 software.

Overview :

The experiment was done in kit mood.

So The inputs and commands were given by keyboard and display were shown in monitor of the kit.

Machine codes were collected from the emu8866 software.

Program - 1

```
MOV AX,1234H  
MOV BX ,4321H  
ADD AX , BX
```

Assembly Language	Machine Code	Calculation
MOV AX , 1234H	B8 34 12	AX = 1234H
MOV BX , 4321H	BB 21 43	BX = 4321H
ADD AX , BX	03 C3	AX = 5555H

MOV is a function to assign a data or address to different address or memory
ADD also assign value to register but with a arithmetic operation of addition.

Output :

```
AX=5555H  
BX=4321H
```

Discussions : In this experiment we learnt basic operating of mda 8086 of machine code and some arithmetic operation also in kit mode. We learnt little basic programming of assembly programming language also. As the MDA-8086 kit output were matched with the emu8086 software, it can be said the experiment was successful.